

20250818EB_MN (TDP43, seahorse)

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Status: In progress

Entry Created On: 09 Sep 2025 04:15:47 UTC

Entry Last Modified: 15 Oct 2025 00:19:16 UTC

Export Generated On: 01 Sep 2026 18:34:56 UTC

2025年8月18日 星期一

DO NOT use 24h time point for this experiment (0.15mM glucose 24h is weird)

split iPSCs into a full 6-well plate on the Friday before differentiation

differentiate 5 of 6-wells with 10mL T2 in cell-repellent plates

keep some iPSCs

2025年8月20日 星期三

T2 -> T3

2025年8月22日 星期五

T3 -> T4 (~14 mL for over-weekend)

2025年8月25日 星期一

T4 -> T5

2025年8月27日 星期三

T5 -> T6

2025年8月29日 星期五

T6 -> T6 (21mL until Tuesday)

2025年9月2日 星期二

T6 -> T7

Dissociation

1. Coat 0.01% Poly-L-Ornithine (PLO) Solution with sodium borate (1:100) for 4-5 hrs
2. Remove the PLO and wash the plate with PBS once
3. Coat the plate with laminine for 1hr.
4. Collect EBs in 50mL falcon and spin them down with 200rcf, 1min. Remove the supernatent (can leave some liquid to avoid remove EBs).
5. Add 2mL accutase in 15mL falcon
6. Put the tube into 37C water bath. Flick the cells every 5 min until the accutase went cloudy (~45min this time). (There are many cells attached on the well of 15mL falcon. Will use 50mL next time)

7. pipette the accutase gently and slowly around 20-30 times to break the EBs by checking if there are still visible EBs.
(Most of the EB could not be dissociated this time, so I take the accutase without visible EBs for the further process and put the undissociated EB to the 2nd round)
8. Add 2x volumn of Heat Inactivated FBS into the tube and pipette it with 5mL serological pipette gently.
9. Add 2x volumn of CTWM and pipette it with 5mL serological pipette gently.
10. Filter the cell with 40um filter into a new 50mL tube
11. Spin the cell down at 200rcf, 3.5min (try faster this time). Remove the supernatent.
12. Add 1ml of T7 to resuspend the cells thouroughly to avoid clusters.
13. Add more T7 (~5mL) to resuspend the cells and count it.
14. Processed the second batch.
15. Plate the cells with T7. (350-500K/well for 12well plate; 1-1.5M/well for 6-well plate)

cell info ^

	A	B	C
1	cell	passage	# of wells
2	WT11	P2	6

Table1 ^

	A	B
1	run1	volume (mL)
2	accutase	4
3	HI-FBS/CTWM	8
4	run2	*take some undissolved EB
5	accutase	2
6	HI-FBS/CTWM	4

Table3 ✓

	A	B	C	D	E	F	G	H	I
1	conc. (M/mL)	survival	total volume	tota cell count	plate#	well plate	cell count/well (K)	number of wells	volume
2	0.811	87	5	4.055	A	24	225	12	3822
3	0.602	91	5	3.01	B	96	60	72	6115
4	0.7065			7.065					

Plate A ^

	1	2	3	4	5	6
A						
B						
C						
D						

Plate B ^

	1	2	3	4	5	6	7	8	9	10	11	12
A												
B												
C												
D												
E												
F												
G												
H												

2025年9月4日 星期四

T7 -> T7

2025年9月5日 星期五

T7 -> BP T8

2025年9月7日 星期日

1/2 T8 -> 1/2 T8

2025年9月9日 星期二

1/2 T8 -> 1/2 T8

Plate B Seahorse

Treat A and B at 11:30am

2025年9月10日 星期三

Plate B Seahorse

1. Rehydrate the cartridge

2. Turn on the seahorse machine.
3. Prepare drugs from stock as the drug table below with two glucose concentrations.
4. every well wash with 90-100uL media with corresponding glucose concentration
5. add 175uL of media to the cells and add 2.5mM glucose media to all the blank wells.
6. incubate the plate for 1 hour to deplete CO2.
7. add drug to the cartridge as below. Add H2O for all the remaining wells.

drug in cartridge layout			^
	A	B	
1	A (oligo)	B (FCCP)	
2	C (Rot/AA)	D (H2O)	

8. calibrate the plate.

Well1													^
	1	2	3	4	5	6	7	8	9	10	11	12	
A	A			B			C			D			
B	1												
C	2												
D	3												
E	4												
F	5												
G	6												
H													

Conditions			
	A	B	C
1	#	glucose concentration	treat time
2	A	2.5mM	24h
3	B	0.15mM	24h
4	C	2.5mM	1h
5	D	0.15mM	1h
6		1 FCCP 0uM	
7		2 FCCP 0.25uM	
8		3 FCCP 0.5uM	
9		4 FCCP 1uM	
10		5 FCCP 1.5uM	
11		6 FCCP 2uM	

drug calculation per glucose concentration																	
	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
1		stock (uM)	1st dilution factor	total volume for 1st dil.	stock volume	media volume for 1st dil.	conc. after 1st dilution	working (uM)	loading volume	well volume before loaded	loading dilution	loading conc.(uM)	dilution factor from 1st dil.	well# for each condition	total loading volume	volume of 1st dil.	volume of media
2	oligo	10000	10	20	2	18	1000	1.5	25	175	8	12	83.3333333333	36	950	11.4	938.6
3	FCCP							0	25	200	9	0	NA	6	200	0.0	200.0
4								0.25	25	200	9	2.25	222.2222222222	6	200	0.9	199.1
5								0.5	25	200	9	4.5	111.1111111111	6	200	1.8	198.2
6								1	25	200	9	9	55.5555555555	6	200	3.6	196.4
7								1.5	25	200	9	13.5	37.037037037	6	200	5.4	194.6
8								2	25	200	9	18	27.7777777777	6	200	7.2	192.8
9	Rot/AA	10000	10	20	1	9	1000	0.5	25	225	10	5	200	36	950	4.8	940.5
10							46										3060.2
11																	3106.2

2025年9月11日 星期四

Treat 24h time point at 16:15

2025年9月12日 星期五

Treat all cells with SPY595 1:3000 at 11:10am for row A,B
row C,D: treat with the drug

Table2



	A	B	C	D	E
1		well#	V/well	total V	SPY595 (1:3000)
2	2.5mM glucose	3	400	1200	0.4
3	0.15mM glucose	3	400	1200	0.4
4	BPT8	4	400	1600	0.5

JFX650

stock:1mM

working:1nM

1. dilute 1mM JFK650 1:400 (0.5uL in 200uL) in no-glu media
2. add 1:2500 into 2.5mM or 0.15mM glucose media

For A2,3 and B2,3:

1. prepare 0.5uL SPY595 in 200uL BP T8 as above.
2. add 50uL into 350uL original media and remove the extra media
3. add the media back to the well at around 11:10am.

For A1 and B1 (24h treatment):

1. dilute SPY595 1:10 with fresh media containing different [gluc]
2. add 1.3 uL into 400uL of old media which was already inside the well (removed extra media) (at around 11:30)
3. pipette up and down
4. 1h prior to the imaging time
5. dilute 1:10 1/400 JFX with old 2.5mM or 0.15mM glucose media
6. add 1.5 uL 1/4000 JFX into the well.

For row 7 and row 8: (treat both SPY595 and JFX650 for 4h)

1. dilute SPY595 1:10 with media with different [gluc]
2. wash cells with media with JFX650
3. add 1.3uL 1/10 SPY595 into 400uL media
4. add back into the well

Well2 ^

	1	2	3	4	5	6
A	1	3	5			
B	2	4	6			
C	7A	7B	7C			
D	8A	8B	8C			

Conditions1 ^

	A	B	C	D
1	#	glucose concentration	treat time	
2	1	2.5mM	24h	
3	2	0.15mM	24h	
4	3	2.5mM	1h	
5	4	0.15mM	1h	
6	5	0.15mM+DMSO	1h	
7	6	0.15mM+150nM Verdinexor	1h	
8	7A	2.5mM	4h	2
9	7B		4h	0.5
10	7C		4h	0.07
11	8A	0.15mM	4h	2
12	8B		4h	0.5
13	8C		4h	0.07

drug prep (800uL/condition)											
	A	B	C	D	E	F	G	H	I	J	K
1		stock (mM)	working (uM)	dilution	1st dilution	0.07	0.5	2	total volume for dilution 1/condition	total volume for dilution 1	stock volume for dilution 1
2	MG132	10.5	2	5250	100	0.5	3.8	15.2	19.580952381	100	1.0
3	Verdinexor	10	0.15	66666.666666667	200				2.4	200	1.0

Treatment time (9/12/2025)									
	A	B	C	D	E	F	G	H	I
1	#			treat time	JFX650 ONLY	image time (1h)	image time (4h)	retreat JFX650	reimage
2	1	2.5mM glucose 24h		9/11 16:15	15:00	15:58			
3	2	0.15mM glucose 24h		9/11 16:15	15:00	16:10		16:47	17:53
4	3	2.5mM glucose 1h		13:00		14:15	17:13		
5	4	0.15mM glucose 1h		13:15		14:23	17:19		
6	5	0.15mM glucose + DMSO		13:30		14:32			
7	6	0.15mM glucose + 150nM Verdinexor		13:45		14:45			
8	7	2.5mM glucose + MG132	A	12:10		16:22			
9			B	12:11		16:34			
10			C	12:12		16:42			
11	8	0.15mM glucose + MG132	A	12:30		16:53			
12			B	12:31		16:59			
13			C	12:32		17:08			

*note: the Halo signal today is not strong, use 30% laser power

*B1 had very low signal -> did I treated JFX650? --> checked the signal between SPY595 and Halo dye: it's quite different signal

But, after re-image the signal is much higher and closer to other images

TDP43 O.C. , 60X, 600x900, 16-bit					
	A	B	C	D	E
1	channel	bit	power (%)	exposure time	note
2	561	16	30	300ms	
3	640	16	30	300ms	



fileid_treat							
	A	B	C	D	E	F	G
1	image	treatment	drug	drug_conc entration	time	media	Halo_tim e
2	MN_TDP43Halo_60x_0001	2.5mM glu			1h	full	1h
3	MN_TDP43Halo_60x_0002	0.15mM glu			1h	full	1h
4	MN_TDP43Halo_60x_0003	0.15mM glu	DMSO		1h	full	1h
5	MN_TDP43Halo_60x_0004	0.15mM glu	Verdinexor 150nM		1h	full	1h
6	MN_TDP43Halo_60x_0005	2.5mM glu			24h	full	1h
7	MN_TDP43Halo_60x_0006	0.15mM glu			24h	full	1h
8	MN_TDP43Halo_60x_0007	2.5mM glu	MG132	2	4h	full	4h
9	MN_TDP43Halo_60x_0008	2.5mM glu	MG132	0.5	4h	full	4h
10	MN_TDP43Halo_60x_0009	2.5mM glu	MG132	0.22	4h	full	4h
11	MN_TDP43Halo_60x_0010	0.15mM glu	MG132	2	4h	full	4h
12	MN_TDP43Halo_60x_0011	0.15mM glu	MG132	0.5	4h	full	4h
13	MN_TDP43Halo_60x_0012	0.15mM glu	MG132	0.22	4h	full	4h
14	MN_TDP43Halo_60x_0013	2.5mM glu	MG132	0	4h	full	4h
15	MN_TDP43Halo_60x_0014	0.15mM glu	MG132	0	4h	full	4h
16	MN_TDP43Halo_60x_0015	0.15mM glu	reimage		24h	full	1h



2025年9月14日 星期日

For 24h timepoint, the result from original one and reimaged one is quite different
--> do not include the 24h time point of this experiment

